

CHRISTIAN MACION

Quantitative Research Analyst · alphaapexresearch@gmail.com · [phone] · [location]

Systematic-strategy researcher who builds, tests, and honestly validates strategies under a pre-registered, multiple-testing-aware framework — statistical validation, transaction-cost realism, and research data engineering.

EXPERIENCE

Quantitative Research Analyst — 19V Capital (formerly Alpha Apex Quant) · 2026 *Systematic strategy-research desk; reporting to Portfolio Manager Evan Ferioli*

- Constructed a **diversified crypto book at annualized Sharpe 1.50** from two independently-validated, near-uncorrelated edges ($\rho = 0.019$; legs 0.97 / 1.04) — passing the desk's portfolio-lift gates, with the Sharpe gain driven by variance reduction (true diversification).
 - Researched **40+ systematic strategy families** across 5 asset classes (equity-index, crypto, energy, metals, agriculture) under a **31-gate statistical filter**; shipped 1 to paper-shadow and advanced 6+ to live forward out-of-sample.
 - Validated **directional and volatility edges** — a post-US-close session effect (Sharpe 0.97; cross-asset random-hour null $z = +4.25$), a volume-breakout continuation (+35.7 bps/fire, $t = 3.01$), and an implied-vol $\rightarrow$ realized-vol forecaster (Spearman IC +0.66, Newey-West t up to +6.7, era-stable).
 - Implemented modern **selection-bias statistics from scratch in numpy** — Deflated Sharpe Ratio, Probability of Backtest Overfitting (CSCV), Minimum Backtest Length — plus block-bootstrap CIs, random-timing nulls, walk-forward, and Monte-Carlo suites.
 - Built **10+ point-in-time data pipelines from scratch** on free public sources (\$0) and an **automated monthly forward-OOS monitoring pipeline** (data pull $\rightarrow$ S3 $\rightarrow$ frozen-spec evaluation $\rightarrow$ ledger) — production-style ML-ops.
 - **Caught and documented three of my own false positives in one session** (a roll-splice artifact that collapsed an apparent $t = 3.0$ edge to $t = 0.3$; an overlap inflation; two pre-registered de-selection sign-flips), banking each as enforced methodology.
 - Designed a **cost-routed multi-agent AI research architecture** (11 agents + a 22-module skills library); authored an institutional operating standard + 16 studies/teaching workbooks adopted across the desk.
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SKILLS

Programming: Python (numpy / pandas / matplotlib / boto3) · from-scratch statistical computing · Git · AWS S3 · Databento (CME L1/L3) · launchd **Methods:** pre-registration & multiple-testing control · block bootstrap · walk-forward · Monte-Carlo · Deflated Sharpe / PBO / MinBTL · cointegration · regime detection · variance-risk-premium · Kelly sizing **Asset classes:** equity-index · crypto (incl. CME micros) · energy · metals · agriculture · prediction markets